

AFD Thermosensory Neurons Mediate Tactile-Dependent Locomotion Modulation in *C. elegans*

Reviewed Preprint

v1 • April 11, 2025

Not revised

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eLife Assessment

This study presents **useful** findings on the role of AFD thermosensory neurons in locomotory behaviours. The study appears **solid** with respect to parsing out the non-thermosensory role of AFD and also brings to light the role of AFD and AIB (linked through electrical synapses) in tactile-dependent locomotory modulation.

<https://doi.org/10.7554/eLife.106496.1.sa3>

Abstract

Sensory neurons drive animal behaviors by detecting environmental stimuli and relaying information to downstream circuits. Beyond their primary roles in sensing, these neurons often form additional synaptic connections outside their main sensory modality, suggesting broader contributions to behavior modulation. Here, we uncover an unexpected role for the thermosensory neuron AFD in coupling tactile experience to locomotion modulation in *Caenorhabditis elegans*. We show that while AFD employs cGMP signaling for both thermotaxis and tactile-dependent modulation, the specific molecular components of the cGMP pathway differ between these two processes. Interestingly, disrupting the dendritic sensory apparatus of AFD, which is essential for thermotaxis, does not impair tactile-based locomotion modulation, indicating that AFD can mediate tactile-dependent behavior independently of its thermosensory apparatus. In contrast, ablating the AFD neuron eliminates tactile-dependent modulation, pointing to an essential role for AFD itself, rather than its sensory dendritic endings. Further, we find tactile-dependent modulation requires the AIB interneuron, which connects AFD to touch circuits via electrical synapses. Removing innexins expressed in AFD and AIB abolishes this modulation, while re-establishing AFD-AIB connections with engineered electrical synapses restores it. Collectively, these findings uncover a previously unrecognized function of AFD beyond thermosensation, highlighting its influence on context-dependent neuroplasticity and behavioral modulation through broader circuit connectivity.

Introduction

Adaptive behavioral responses require animals to interpret sensory signals in the context of prior experiences, enabling effective responses to changing environmental conditions. For example, rodents associate contextual cues, such as odors and colors, with rewards or punishments, leading to context-dependent adjustments in approach or avoidance behaviors [1,2]. Disruptions in the biological processes underlying this plasticity, as seen in neurological and psychiatric disorders, often result in impaired behavior modulation based on prior sensory experiences [3–7]. Recognizing the critical role of context-dependent behavioral modulation for survival and well-being, ongoing research focuses on uncovering the neuroplastic mechanisms responsible for these adaptive processes.

Studies across diverse organisms reveal that animals, regardless of nervous system complexity, integrate prior sensory experiences to shape behavior [8–15]. In the nematode *C. elegans*, which has only 302 neurons, locomotion strategies are modified by prior exposure to stimuli such as odors, touch, light, and starvation, demonstrating robust adaptive behaviors [16–23]. The wide range of adaptive behaviors, combined with the well-mapped *C. elegans* nervous system and abundant genetic tools, have empowered the discovery of fundamental circuit mechanisms and molecular pathways underlying context-dependent behavioral modulation [24–28].

The *C. elegans* sensory system comprises 60 ciliated sensory neurons specialized in detecting chemical, thermal, mechanical, and olfactory stimuli [29,30]. Each sensory neuron has a dendritic apparatus that converts specific environmental signals into neuronal activity, which drives signal flow through downstream networks of interneurons and motor neurons, ultimately shaping behavior [31–33]. Among these, AFD, a pair of bilaterally symmetric left-right neurons, serves as the primary thermosensory neurons required for thermotaxis (movement toward the a preferred temperature) [34]. The dendritic endings of AFD, enriched with microvilli, serve as the primary site for thermal detection. Even when detached from the cell body, these dendritic endings retain calcium responsiveness to temperature stimuli [35]. Moreover, genetically disrupting these sensory endings abolishes the Ca^{2+} response in AFD cell bodies to temperature changes and eliminates thermotaxis, underscoring their essential role in thermal sensing [36–38].

The second messenger cyclic guanosine monophosphate (cGMP) plays a central role in modulating AFD thermosensory activity. Temperature changes in AFD neurons trigger intracellular Ca^{2+} dynamics, with Ca^{2+} levels rising during warming and falling during cooling [35,39,40]. This response relies on cGMP synthesis mediated by the guanylyl cyclases (GCs) *gcy-8*, *gcy-18*, and *gcy-23*, which have overlapping but distinct roles. Specifically, *gcy-8* is essential for generating thermally evoked Ca^{2+} responses in AFD [41,42], whereas *gcy-18* and *gcy-23* act synergistically to sustain temperature sensing [41,43]. The cGMP produced by these cyclases activates cyclic nucleotide-gated (CNG) channels, comprised of alpha-subunits TAX-4 and beta-subunits TAX-2, resulting in increases in intracellular Ca^{2+} . Mutations in either *tax-4* or *tax-2* abolish thermally evoked Ca^{2+} responses in AFD and disrupt thermotaxis, underscoring the importance of cGMP signaling for AFD-mediated thermosensation [40].

AFD transmits thermal signals to several downstream interneurons, including AIY, AIZ, and AIB, which process this information to guide temperature-dependent behaviors. Thermotaxis in *C. elegans* encompasses three categories: movement toward warmer temperatures (thermophilic), movement toward cooler temperatures (cryophilic), and movement within a preferred temperature range (isothermal) [44,45]. Each behavior type depends on distinct AFD synaptic partners: AIY is critical for thermophilic and isothermal behaviors [46,47]; AIZ is specific for

cryophilic behavior [48–50]; and AIB also supports thermophilic behavior, though its role is less critical than that of AIY [51]. These findings demonstrate the functional specificity of synaptic partners of AFD neurons in shaping distinct temperature-dependent behaviors.

In this study, we demonstrate that AFD plays a novel role in tactile-mediated, context-dependent locomotion modulation, thereby expanding its function beyond temperature sensing. The mechanisms underlying this role differ markedly from those involved in thermosensation. First, context-dependent modulation requires distinct cGMP signaling components. Second, while the microvilli-rich sensory endings of AFD are essential for thermosensation, they are dispensable for context-dependent modulation. Third, this modulation depends on mechanosensory neurons and on electrical synapses between AFD and the AIB interneuron, which connects to mechanosensory circuits. Finally, disruption of *inx-7* and *inx-10* genes, encoding gap junction protein innexins in AFD and AIB, impairs context-dependent locomotion modulation. Remarkably, restoring AFD–AIB connectivity via engineered Cx36 electrical synapses rescues this function. Collectively, our findings uncover an unconventional mechanism by which AFD mediates behavior modulation based on tactile experience.

Results

C. elegans displays distinct locomotion in identical environments after exploring different physical structures

To investigate the impact of prior experience on *C. elegans* behavior, we first allowed worms to explore chambers with distinct pillar designs (exploration zone, **Fig. 1A**) before recording their locomotion in identical assay zones (red dashed-line boxes, **Fig. 1A**). We hypothesized that if behavior were solely driven by immediate sensory input, worms in identical assay zones would exhibit similar locomotion. Conversely, if prior experience, such as exposure to different physical features in the exploration zones, influences behavior, then worms should display distinct locomotor patterns even in the identical assay zones. To test these hypotheses, we employed two types of microfluidic chambers: uniform and binary (**Fig. 1A**). Both chamber types include an identical assay zone. Once worms enter this zone, their immediate environment is the same regardless of the chamber. However, before reaching the assay zone, worms explore surrounding areas with pillar designs that differ in size and spacing. This design allowed us to assess how prior exposure to different physical contexts impacts locomotion in an otherwise identical setting.

Worms were allowed to explore the chambers for 60 minutes before their behavior was recorded in the assay zones. We found that worms in the uniform chamber assay zone exhibited locomotor behavior that differed from that of worms in the binary chamber assay zone (**Fig. 1B–F**). Specifically, in the uniform chamber assay zone, worms moved approximately 40% faster (**Fig. 1B, F**), spent about 20% less time idle (**Fig. 1C, F**), reversed direction around 20% less frequently (**Fig. 1D, F**), and turned about 30% more often (**Fig. 1E, F**) than worms in the binary chamber assay zone. These findings demonstrate that locomotion adjustments are shaped by past sensory experience rather than solely by immediate environmental cues. That worms exhibited different behaviors in identical assay zones highlights the flexible, context-dependent nature of *C. elegans* behavior.

Guanylyl cyclase gene *gcy-18* is required for context-dependent locomotion modulation in chambers

To identify mechanisms underlying context-dependent locomotion adjustments, we employed a previously reported preference assay that uses microfluidic arenas to assess the ability of worms to select a preferred space [52]. Because worms must adjust their locomotion based on their surroundings to remain within a preferred area, we hypothesized that mutations abolishing this

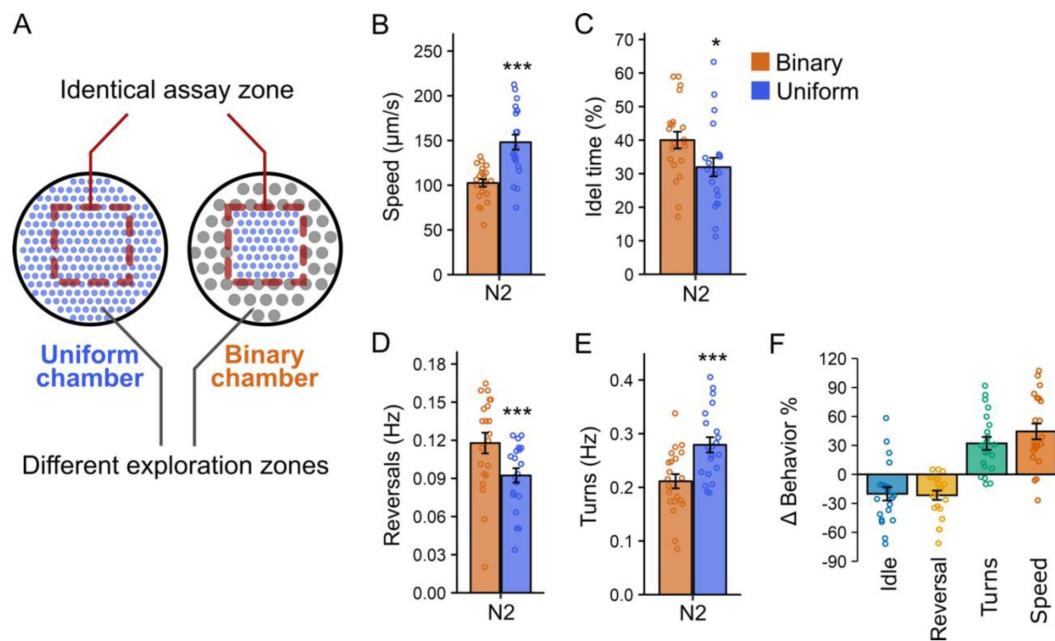


Figure 1.

***C. elegans* displays distinct locomotion in identical environments after exploring different surrounding areas.**

(A) Schematic of two microfluidic chamber designs—uniform and binary—used for behavioral assays. Both chambers feature an identical assay area (red dashed-line boxes) where locomotion was analyzed, while the exploration zones contain distinct PDMS pillar patterns for worm exploration. (B–E) Bar graphs showing locomotion properties of wild type N2 worms in the assay zones of both chamber types after 1 hour of exploration. Data from binary chambers are shown in orange and from uniform chambers in blue. Shown are (B) locomotion speed, (C) percentage of time spent idle, (D) reversal frequency, and (E) turning frequency. Error bars represent standard error of the mean (SEM); each data point represents the mean behavior value of worms within a chamber. (F) Percent changes in locomotion properties, calculated from data in panels B–E and normalized to the mean value obtained from binary chambers (see Methods), are presented as mean $\pm$ SEM. Statistical significance was determined using an unpaired Student's t-test (***: $p < 0.001$, *: $p < 0.05$).

preference would reveal key mechanisms underlying context-dependent adjustments in the uniform and binary chambers. Accordingly, we screened for mutant worms that failed to choose their preferred area, aiming to identify the genetic mechanisms mediating locomotion adjustments in response to different physical structures. Given the significant role of cGMP in neuronal plasticity and behavior modulation, we performed a small-scale genetic screen using loss-of-function mutants for 24 genes encoding guanylyl cyclase (GC), the enzyme responsible for cGMP synthesis. Using the preference assay, we identified two putative null mutants, *gcy-18(gk423024)* and *gcy-12(gk142661)*, that failed to display a spatial preference (**Supp Fig. 1**). We then compared the locomotion properties of *gcy-18* and *gcy-12* mutants in both uniform and binary chambers. Our results showed that *gcy-18* mutants failed to execute context-dependent adjustments, displaying similar locomotion rates in both chamber types. Specifically, *gcy-18* mutants exhibited a significantly reduced Δspeed ($3 \pm 5\%$) compared to wild type ($35 \pm 7\%$, $p < 0.001$, **Fig. 2A**). In contrast, *gcy-12* mutant worms exhibited normal speed adjustment, with Δspeed values comparable to wild type (**Fig. 2A**). This indicates that *gcy-12* is not required for the modulation of locomotion speed. In comparison, the failure of *gcy-18* mutants to adjust their speed suggests that *gcy-18* plays a critical role in context-dependent behavioral adjustment. Together, these results demonstrate that among the guanylyl cyclases, *gcy-18* is essential for the context-dependent modulation of locomotion.

To confirm that the defects observed in *gcy-18* mutants were not due to background artifacts, we introduced a single-copy transgene that drives *gcy-18* cDNA expression under the control of a *gcy-18* promoter into *gcy-18(gk423024)* mutants to test whether reintroducing a functional copy of *gcy-18* could rescue the behavioral defects. Indeed, expression of the transgene restored locomotion adjustments (Δspeed : $20 \pm 6\%$) to levels closer to those observed in wild type animals ($35 \pm 7\%$, $p = 0.4$, **Fig. 2B**), confirming the crucial role of *gcy-18* in mediating these adjustments based on the experience of exploring physical environments. The identification of *gcy-18* was particularly intriguing, as its expression is restricted to a single pair of neurons, the thermosensory AFDs. Given that our assay chambers lacked thermal gradients, this finding suggests a novel role for *gcy-18* and AFD neurons in mediating locomotion adjustments in response to the exploration of surrounding physical structures.

To further investigate the role of AFD-specific guanylyl cyclases in context-dependent locomotion modulation, we examined *gcy-8*, which is essential for generating thermally evoked Ca^{2+} responses in AFD [41,42]. In thermal responses, disrupting *gcy-8* alone is sufficient to impair AFD function, whereas disruption of *gcy-18* alone does not [41,43], indicating the critical role of GCY-8 in AFD thermosensation. If *gcy-18* influenced behavioral adjustment in microfluidic chambers solely through its role in thermosensation, we would expect that a loss-of-function mutation in *gcy-8* would have a similar or even more disruptive impact on locomotion modulation. However, *gcy-8* mutant worms exhibit little change in their ability to adjust locomotion speed (Δspeed : $29 \pm 7\%$), with Δspeed values similar to that observed in wild type worms ($p = 0.96$, **Fig. 2C**). These results indicate that while *gcy-8* is critical for thermosensation, its mutation does not affect context-dependent locomotion modulation, suggesting that independent cGMP signaling mechanisms govern AFD neuronal activity in thermosensation and context-dependent behavior in microfluidic chambers (**Fig. 2D**).

Distinct CNG channels mediate context-dependent locomotion modulation and thermosensation

The role of guanylyl cyclase in cGMP synthesis prompted us to investigate whether cyclic nucleotide-gated (CNG) channels are involved in context-dependent locomotion modulation. CNG channels, composed of α and β subunits, open upon binding cyclic nucleotides (e.g., cGMP and cAMP), allowing Ca^{2+} influx [53–56] to control signal transduction and neural plasticity across species [57–64] (**Fig. 3A**). In *C. elegans*, mutations in either the *tax-4* or *tax-2* genes, which encode the α and β subunits of CNG channels respectively, impair thermotaxis behavior

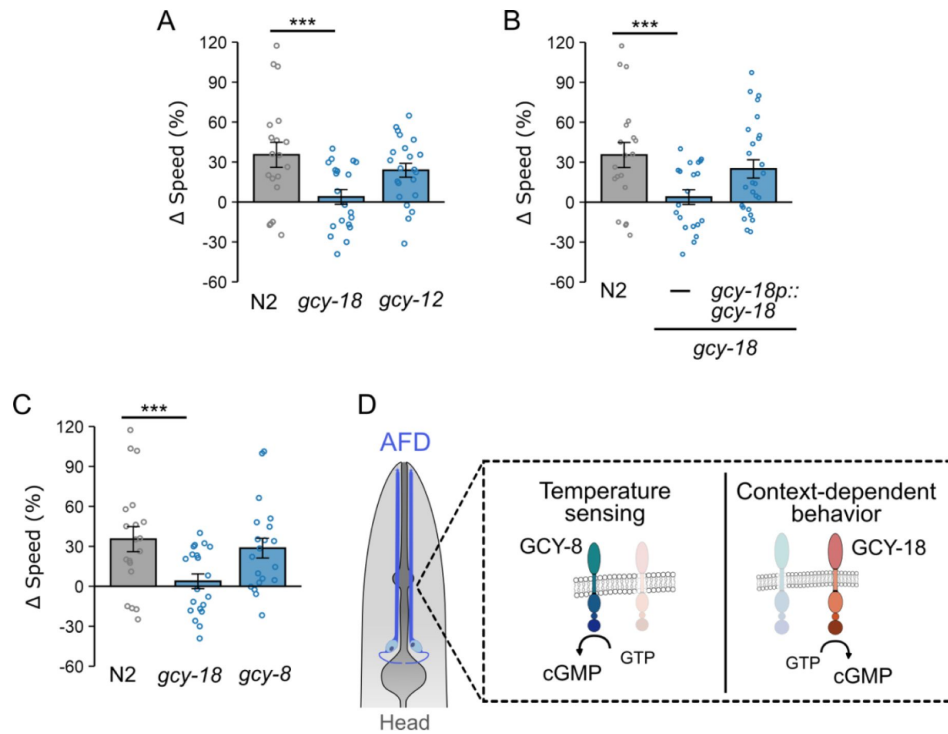


Figure 2.

Guanylate cyclase gene *gcy-18* functions in AFD to mediate context-dependent locomotion modulation.

(A) Speed differences (Δ speed) between uniform and binary chambers of wild type N2, *gcy-18* and *gcy-12* mutant worms. Δ Speed was calculated as the percent change in locomotion rates in uniform chambers, relative to the mean of values obtained in binary chambers (see Methods). Mutant worms lacking *gcy-18* failed to demonstrate context-dependent adjustments in locomotion speed, as indicated by a low Δ speed value, whereas *gcy-12* mutants behaved similarly to wild type worms. (B) A single-copy transgene expressing *gcy-18* cDNA under the control of the *gcy-8* promoter in AFD restores context-dependent locomotion modulation. (C) Disruption of *gcy-8* does not impair context-dependent locomotion modulation. (D) Schematic illustrating the distinct roles of guanylate cyclase genes in AFD sensory neurons: *gcy-8* is required for thermosensation but not for context-dependent locomotion modulation, whereas *gcy-18* is essential for context-dependent locomotion modulation and plays only a moderate role in thermosensation. Data are presented as mean $\pm$ SEM. Each data point represents the mean behavior of worms within a single chamber. Statistical significance was determined using one-way ANOVA followed by a Tukey–Kramer post hoc test (***: $p < 0.001$).

[47,65] and disrupt AFD neuron responses to temperature changes [40,66]. To determine their role in context-dependent locomotion modulation, we examined *tax-4(p678)* and *tax-2(p694)* mutant worms using our microfluidic chamber assay.

We found that *tax-2* is essential for context-dependent locomotion modulation, as *tax-2* mutants did not exhibit differences in speed between chamber types (**Fig. 3B, C**). In contrast, *tax-4* mutants displayed an enhanced locomotion adjustment compared to wild type worms ($p < 0.01$; **Fig. 3C**)—a striking difference from the lack of speed changes seen in *tax-2* mutants. Furthermore, *tax-4* mutants exhibited significantly increased basal locomotion rates in both chambers (**Fig. 3B**), suggesting that alterations in basal locomotion do not interfere with experience-dependent movement adjustments.

Consistent with the idea that basal and experience-dependent locomotion are regulated independently, we found that wild type worms cultivated at either 23°C or 17°C and then transferred to 20°C during chamber exploration exhibited significant changes in basal locomotion speed but had little corresponding effect on the extent of context-dependent modulation (**Suppl. Fig. 2A**). Specifically, temperature shifts were positively correlated with basal locomotion rates (binary chambers: $r = 0.52$, $p < 0.0001$; uniform chambers: $r = 0.48$, $p < 0.0001$; **Suppl. Fig. 2B**), yet no significant correlation was observed between temperature shifts and Δspeed ($r = -0.03$, $p = 0.22$; **Suppl. Fig. 2C**), reinforcing the idea that these two processes operate independently.

Together, these observations indicate that TAX-2 is critical for integrating prior experience into locomotion modulation, while TAX-4 primarily regulates basal locomotion. Despite both pathways engaging cGMP signaling, their functional independence suggests distinct regulatory mechanisms for basal locomotion and context-dependent adjustments.

Given that four other genes (*cng-1*, *cng-2*, *cng-3* and *cng-4*) are predicted to encode CNG channels in *C. elegans*, we investigated their role in locomotion modulation in microfluidic chambers [67]. Due to abnormal development in *cng-1* mutants, we were unable to assess their locomotion; however, we were able to evaluate mutants of the other three genes. We found that *cng-3(gk541751)* mutant worms failed to adjust locomotion speed in the microfluidic chambers, while *cng-2(tm4267)* and *cng-4(gk195496)* mutants exhibited normal modulation comparable to wild type worms (**Fig. 3D**). The identification of CNG-3 was particularly intriguing, as recent studies suggest that it mediates changes in AFD neuronal activity based on prior temperature experiences [68,69], indicating that CNG-3 contributes to neural plasticity rather than direct stimulus detection. Our results further support this view by showing that CNG-3 plays a broader role in locomotion modulation in environments lacking thermal gradients, which suggests its involvement in neuronal plasticity beyond the thermosensory modality.

Next, we determined whether AFD is the site where CNG-3 functions in context-dependent locomotion modulation. To do so, we introduced single-copy transgenes expressing *cng-3* cDNA under the control of two AFD-specific promoters, *srtx-1Bp* and *gcy-18p*, into *cng-3(gk541751)* mutant worms. Regardless of the promoter used, AFD-specific expression of *cng-3* restored the locomotion speed modulation (Δspeed : $47 \pm 8\%$ for *srtx-1Bp::cng-3* and $40 \pm 6\%$ for *gcy-18p::cng-3*) to levels similar to wild type (**Fig. 3E**). These results demonstrate that AFD neurons are the site of action for CNG-3 in mediating context-dependent locomotion modulation. Overall, our findings elucidate an unexplored role for AFD beyond thermosensation, revealing that components of a cGMP signaling pathway enable AFD to modulate behavior in an experience-dependent manner.

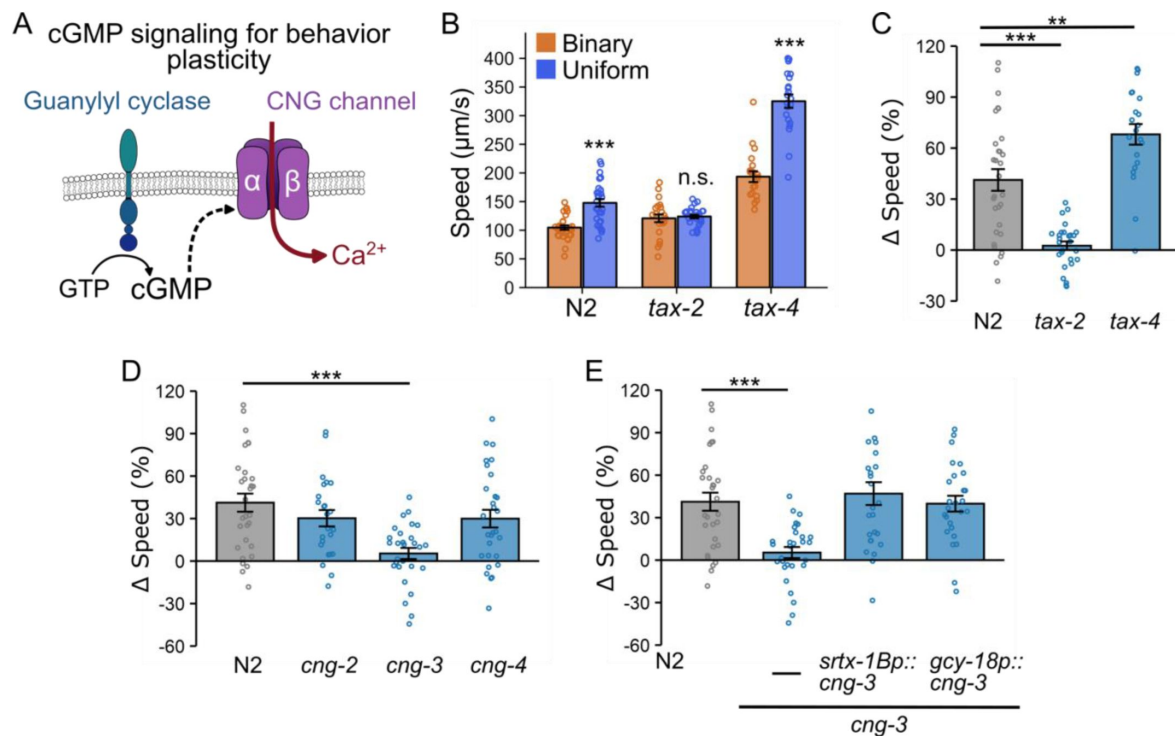


Figure 3.

Cyclic nucleotide-gated (CNG) channel subunits TAX-2 and CNG-3 are required for locomotion modulation, but TAX-4 is not.

(A) Schematic representation of CNG channel function. CNG channels are activated by cyclic nucleotides (such as cGMP) and mediate Ca²⁺ influx, thereby influencing neuronal activity and animal behavior. (B) Locomotion speed (μm/s) of N2, *tax-2*, and *tax-4* mutant worms in binary (orange) and uniform (blue) chambers. The *tax-2* mutants exhibit identical locomotion rates in both chamber types, whereas worms lacking *tax-4* display accelerated basal locomotion while still preserving context-dependent locomotion modulation. (C) Speed differences (Δspeed) for N2, *tax-2* mutant, and *tax-4* mutant worms. The *tax-2* mutants failed to modulate locomotion rates in a context-dependent manner, whereas *tax-4* mutations enhanced modulation. (D) Assessment of *cng-2*, *cng-3*, and *cng-4* roles in locomotion adjustments. The *cng-3* mutation abolishes locomotion modulation. (E) Single-copy transgenes expressing *cng-3* cDNA under two different AFD-specific promoters restore context-dependent locomotion adjustments. Data are presented as mean ± SEM. Each data point represents the mean behavior of worms within a single chamber. Statistical significance was determined using an unpaired Student's t-test to compare speeds between uniform and binary chambers (panel B) and one-way ANOVA followed by a Tukey-Kramer post hoc test to compare Δspeed across strains (panels C-E) (*: $p < 0.05$; **: $p < 0.01$, and ***: $p < 0.001$).

AFD sensory microvilli are dispensable for locomotion modulation, but AFD neurons are required

The finding that cGMP signaling genes (such as *gcy-18* and *cng-3*) in AFD neurons modulate context-dependent locomotion prompted us to investigate whether the AFD thermosensory apparatus, crucial for temperature detection [36], is also necessary for locomotion modulation in our microfluidic assay. Previous studies have shown that the sensory microvilli at the tip of the AFD dendrite (Fig. 4A) are essential for temperature sensing and thermotaxis [36,38]. For example, mutations in the *kcc-3* gene, which encodes a K⁺/Cl⁻ cotransporter in the amphid sheath glia, disrupt microvilli formation and significantly impair thermotaxis towards their cultivation temperature [37]. Despite these thermotaxis defects, we found that context-dependent locomotion modulation remained unchanged in *kcc-3(ok228)* mutant worms (Δ speed: $33 \pm 4\%$) compared to wild type ($42 \pm 7\%$; $p = 0.28$; Fig. 4B, C). Interestingly, *kcc-3* mutant worms displayed elevated basal locomotion rates in both chamber types (Fig. 4B), a phenomenon also observed in *tax-4* mutants. This further supports our earlier finding that alterations in basal locomotion do not correlate with context-dependent locomotion modulation (Suppl. Fig. 2B, C). Together, these findings suggest that the role of AFD neurons in modulating context-dependent locomotion is distinct from their thermosensory functions and differs from the mechanisms controlling basal locomotion.

To further confirm that the sensory endings of AFD are not required for context-dependent locomotion modulation, we examined *ttx-1(767)* mutant worms, which completely lack microvilli and exhibit more severe thermotaxis defects than those seen in *kcc-3* mutants [38]. Consistent with the notion that AFD has separate roles in context-dependent locomotion and thermosensation, *ttx-1* mutants still display locomotion modulation in microfluidic chambers. However, the extent of modulation in these mutants was slightly reduced and more variable (Δ speed: $29 \pm 10\%$) compared to wild type animals ($42 \pm 7\%$, $p = 0.29$, Fig. 4D). It is worth noting that *ttx-1* mutations disrupt the expression of multiple AFD genes, including those involved in cGMP signaling [70], which may explain the increased variability in locomotion modulation. Nonetheless, despite completely lacking the AFD sensory apparatus, the ability of *ttx-1* mutant worms to adjust their locomotion speed further demonstrates that AFD neurons modulate context-dependent locomotion via mechanisms distinct from those used in thermosensation.

Surprised to find that the AFD sensory apparatus is dispensable for locomotion modulation, we next examined worms with genetically ablated AFD neurons. To achieve targeted AFD ablation, we used a published strain with an integrated transgene expressing caspase protein under the AFD-specific promoter *gcy-8p* [71]. AFD-ablated worms lost the ability to adjust locomotion rates in microfluidic chambers, as indicated by a significant reduction in Δ speed (wild type: $42 \pm 7\%$, AFD ablated: $-2 \pm 7\%$, $p < 0.001$; Fig. 4E). These results demonstrate that although the sensory endings of AFD are not required for locomotion adjustment, the AFD neuron itself is essential.

Context-dependent locomotion modulation requires the MEC-10 mechanosensory channel subunit

We investigated whether mechanosensation is necessary for worms to perform context-dependent locomotion adjustments in microfluidic chambers. This was prompted by previous findings showing that touch sensation is critical for *C. elegans* to select their preferred space through interactions with physical structures [52]. The *mec-10* gene encodes an amiloride-sensitive Na⁺ channel protein required for sensing body touch. Mutations in *mec-10* disrupt mechanotransduction channels in the worms touch-receptor neurons (TRNs), impairing touch sensitivity [72,73]. We found that *mec-10(tm1552)* mutants failed to display locomotion

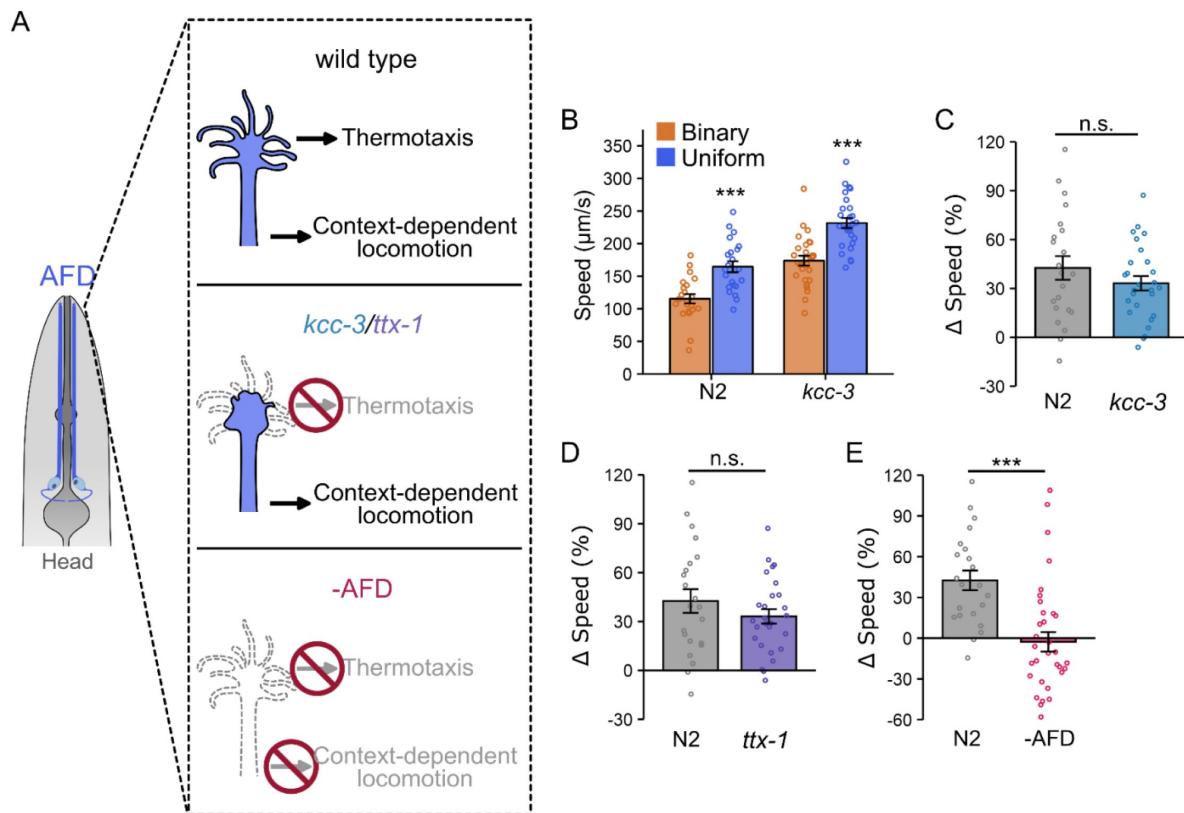


Figure 4.

AFD, but not its sensory endings, is required for context-dependent locomotion modulation.

(A) Schematic representation of AFD function in temperature sensing and locomotion modulation. AFD sensory endings are essential for thermosensation but dispensable for context-dependent locomotion modulation. (B) Locomotion speed (μm/s) of wild type N2 and *kcc-3* mutant worms in the binary (orange) and uniform (blue) chambers. The *kcc-3* mutants preserve context-dependent locomotion modulation while exhibiting increased basal locomotion rates. (C) Speed differences (Δspeed) for N2 and *kcc-3* mutant worms, and (D) for N2 and *ttx-1* mutant worms, respectively. Context-dependent locomotion modulation remains in *kcc-3* and *ttx-1* mutant worms, although they abolish the AFD thermosensory function. (E) Ablation of AFD eliminates the context-dependent locomotion modulation. Data are presented as mean ± SEM. Each data point represents the mean behavior of worms within a single chamber. Statistical significance was determined using an unpaired Student's t-test (n.s. indicates not significant, and ***: $p < 0.001$).

adjustment (Δ speed: $5 \pm 4\%$, **Fig. 5B**). These results show that *mec-10* is crucial for context-dependent locomotion modulation, supporting the hypothesis that locomotion is modulated depending on the worm's tactile experience.

AIB interneurons connecting TRNs and AFD are crucial for tactile-dependent behavioral plasticity

Touch receptor neurons (TRNs) do not directly synapse onto AFD, suggesting that additional neurons are required to bridge them for tactile-dependent locomotion modulation. Examination of the *C. elegans* connectome revealed that AIB serves as the shortest network pathway connecting TRNs to AFD (**Fig. 5A**). This positions AIB as a key candidate for relaying signals from TRNs to AFD. If this is correct, we expect AIB to be necessary for tactile-dependent modulation. To examine this, we quantified the locomotion of AIB-ablated worms in microfluidic chambers. For this, we employed a previously published strain expressing caspase under an AIB-specific promoter (*npr-9p*), resulting in ablation of the AIB neuron pair [74]. We found AIB-ablated worms failed to adjust their speed in chambers, showing a significant reduction in Δ speed ($-1 \pm 5\%$) compared to wild type ($32 \pm 8\%$, $p < 0.01$, **Fig. 5C**). These results demonstrate that the AIB interneurons are crucial for tactile-dependent locomotion modulation.

Mutations in *innexin* genes disrupt tactile-dependent modulation

Electrical synapses between AIB and AFD in *C. elegans* are formed by gap junction proteins called innexins [75]. If these AIB–AFD connections are important for tactile-dependent modulation, then mutations in innexin genes should impair locomotion adjustments in microfluidic chambers. To examine this hypothesis, we investigated loss-of-function mutants of various innexin genes. Because *C. elegans* has 25 innexin genes [76,77], we focused on those known to be expressed in AFD and AIB [78–80], identifying three genes (*inx-4*, *inx-10* and *inx-19*) expressed in AFD, one gene (*inx-1*) in AIB, and one gene (*inx-7*) in both. We found that *inx-10* mutants exhibited a partial but significant decrease in locomotion adjustments (Δ speed: $15 \pm 3\%$) compared to wild type worms ($45 \pm 7\%$, $p < 0.001$, **Fig. 6A**). Similarly, mutations in *inx-1*, *inx-4*, and *inx-7* all lowered Δ speed values, although these differences were not statistically significant relative to wild type worms (**Fig. 6A**). Furthermore, we found that *inx-7; inx-10* double mutants exhibit a lower Δ speed compared to *inx-7* or *inx-10* single mutants (**Fig. 6B**). These findings suggest that disrupting innexins in AFD and AIB impair tactile-dependent locomotion modulation.

Re-establishing AIB-AFD electrical synapses restores tactile-dependent locomotion modulation in *inx-7; inx-10* double mutants

If the tactile-dependent modulation defects in innexin mutants result from disrupted AIB-AFD electrical synapses, then reintroducing a synthetic gap junction specifically between these neurons should rescue locomotion modulation. To test this, we employed a recently developed approach that introduces engineered electrical synapses by expressing the mammalian gap junction protein Cx36 in *C. elegans* neurons. Several studies have shown that engineered Cx36 synapses in *C. elegans* facilitate electrical coupling between neurons, support behavioral modulation, and circumvent chemosensory impairments in damaged neural circuits [81–84]. Based on these findings, we introduced Cx36 into AFD and AIB neurons in *inx-7; inx-10* double mutants (**Fig. 6C**). Remarkably, expression of Cx36 in both AFD and AIB significantly improved locomotion adjustments in *inx-7; inx-10* mutants, increasing Δ speed from $15 \pm 3\%$ to $31 \pm 8\%$ ($p < 0.05$, **Fig. 6D**). These findings demonstrate that engineered electrical synapses between AFD and AIB can restore tactile-dependent locomotion modulation.

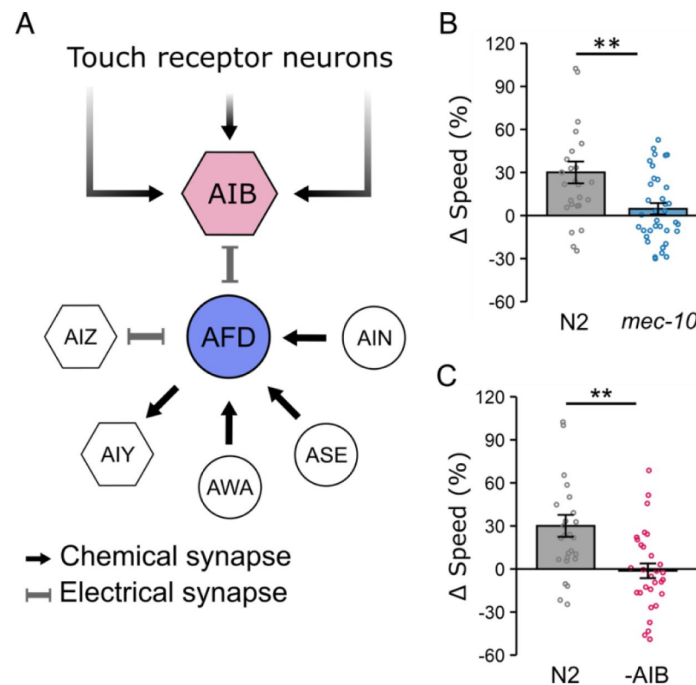


Figure 5.

Context-dependent locomotion adjustments require both the mechanosensory channel subunit MEC-10 and the AIB interneurons that connect AFD to touch circuits.

(A) Schematic representation of neurons synapsing with AFD. AIB provides the shortest pathway linking AFD to touch receptor neurons. (B, C) Speed differences (Δ speed) for N2, *mec-10* mutants, and AIB-ablated (-AIB) worms. Both *mec-10* mutants (B) and AIB-ablated worms (C) fail to modulate locomotion in microfluidic chambers. Data are presented as mean $\pm$ SEM. Each data point represents the mean behavior of worms within a single chamber. Statistical significance was determined using one-way ANOVA followed by a Tukey–Kramer post hoc test (**: $p < 0.01$).

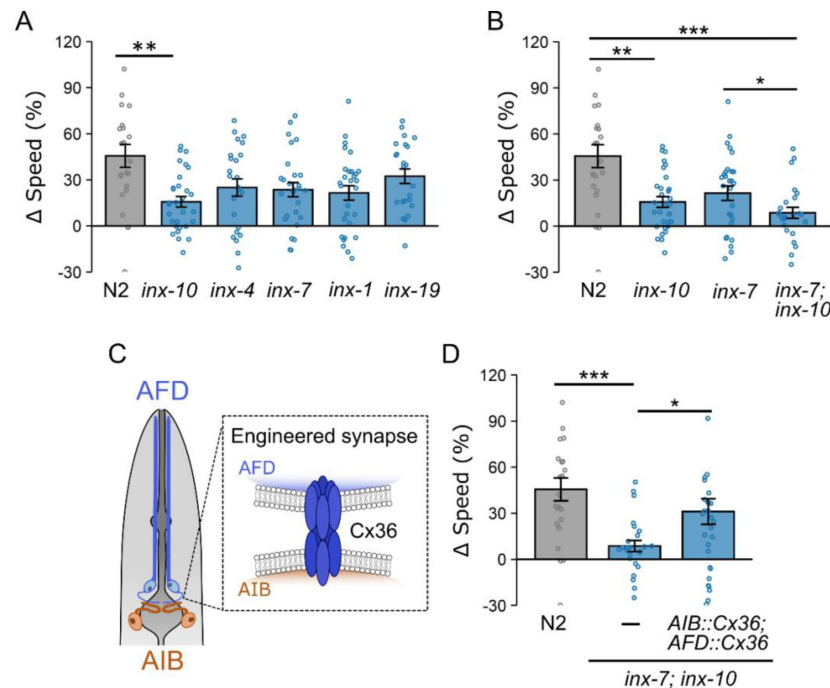


Figure 6.

Tactile-dependent locomotion modulation is disrupted in mutant worms lacking gap junction genes and is restored by engineered Cx36 electrical synapses linking AFD and AIB.

(A) Speed differences (Δ Speed) for *inx-1*, *4*, *7*, *10*, *19* mutant worms. (B) Speed differences for *inx-7; inx-10* double mutants. (C) Schematic illustrating engineered electrical synapses formed by Cx36 when expressed in adjacent neurons. (D) Tactile-dependent locomotion modulation is restored in *inx-7; inx-10* double mutants by expressing Cx36 in AFD and AIB neurons. Data are presented as mean $\pm$ SEM. Each data point represents the mean behavior of worms within a single chamber. Statistical significance was determined using one-way ANOVA followed by a Tukey-Kramer post hoc test (*: $p < 0.05$; **: $p < 0.01$, and ***: $p < 0.001$).

Discussion

Our study reveals that the thermosensory AFD neuron plays an unexpected role in coupling tactile experiences to locomotion modulation in *C. elegans*. We show that: 1) *C. elegans* exhibit distinct locomotion patterns in identical environments, influenced by prior tactile experiences; 2) tactile-dependent behavior modulation requires the AFD neuron but not its thermosensory dendritic endings; 3) thermotaxis and tactile-dependent locomotion modulation rely on distinct sets of cGMP signaling proteins and CNG channel subunits; 4) the AIB interneuron, which connects AFD to mechanosensory circuits, is essential for tactile modulation; and 5) electrical synapses between AFD and AIB are critical for this process. Below, we discuss the implications of these findings.

A thermosensation-independent role for AFD in tactile-dependent behavior modulation

Our findings highlight the remarkable versatility of sensory neurons in shaping behaviors beyond their primary functions. This supports the emerging view that sensory neurons have multifaceted roles in circuit modulation, extending well beyond canonical sensory activities. Such complexity is encoded at multiple levels within neural circuits. In *C. elegans*, for instance, the ASH neuron drives avoidance responses to various noxious stimuli, including hyperosmolarity, nose touch, and volatile repellents, demonstrating polymodal functionality [85–88]. Similarly, the olfactory receptor neuron AWC exhibits stochastic responses to temperature, introducing variability that enhances the flexibility and reliability of thermotactic behavior [89–91]. In contrast to these examples, the role of AFD in tactile-dependent plasticity does not rely on its sensory endings but is mediated through synaptic connectivity. This reveals an unexpected mechanism by which AFD influences context-dependent behavioral plasticity via network-level interactions rather than direct sensory activity.

Distinct cGMP signaling pathways in AFD mediate sensation and plasticity

Our data indicates that the CNG channel α subunits TAX-4 and CNG-3 play distinct roles in AFD function. Previous work has shown that TAX-4 is essential for AFD to detect temperature and generate thermally evoked responses [40], whereas CNG-3, although dispensable for temperature sensing, is crucial for tactile-dependent locomotion modulation. Supporting a broader role in sensory plasticity, CNG-3 is required in AWC neurons for short-term odor adaptation [92] and in AFD for adjusting temperature thresholds based on prior thermal experiences [68]. Our demonstration of the essential role of CNG-3 in tactile-dependent locomotion modulation further suggests that CNG-3 enables context-dependent behavioral plasticity rather than directly mediating stimulus detection.

CNG channels are heteromeric complexes composed of both α and β subunits. Our data, along with previous work, show that TAX-2, the CNG β subunit in *C. elegans*, is required for both tactile-dependent locomotion modulation and thermotransduction [40]. This finding implies that TAX-2 can partner with different α subunits to support sensory detection and neural plasticity. This functional specialization is reminiscent of photoreceptor neurons, where the composition and ratio of CNG channel subunits determine light sensitivity and enable contrast detection under varying light intensities [93–95].

In addition to the diversity of CNG channels, guanylyl cyclases in AFD also exhibit functional specificity. For example, mutations in *gcy-18* alone significantly disrupt tactile-dependent locomotion modulation without substantially affecting temperature sensing [41,43], whereas *gcy-8* is essential for generating thermally evoked Ca^{2+} responses [41,42] yet appears

dispensable for tactile-dependent plasticity. Together, these results show that AFD uses distinct sets of guanylyl cyclases and CNG channels to execute dual roles—precise temperature sensing and context-dependent behavioral modulation.

Differential regulation of context-dependent plasticity and basal locomotion rates in AFD

Our results further suggest that the mechanisms controlling basal locomotion and context-dependent behavioral modulation are not only distinct but also operate independently. Temperature fluctuations significantly affect basal locomotion rates, i.e., higher temperatures accelerate speed, while lower temperatures slow it down [96,97]. However, tactile-dependent locomotion modulation in microfluidic chambers remains robust despite these temperature-induced changes in basal movement. This independence is further supported by our observations in *tax-4* and *kcc-3* mutants, which display elevated basal locomotion rates yet still adjust their locomotion in response to tactile experiences. These findings indicate that the processes governing basal locomotion and context-dependent modulation are controlled by separate, non-interfering mechanisms, even though both pathways engage cGMP signaling.

Electrical synapses bridge distinct circuits for behavioral modulation

AIB-AFD electrical synapses are pivotal for tactile-dependent behavioral plasticity, serving to connect mechanosensory and thermosensory circuits. Our results show that disrupting the electrical connection between AFD and AIB impairs tactile-dependent locomotion modulation, whereas reintroducing engineered synapses restores it. The successful replacement of invertebrate innexins with vertebrate connexins, proteins that perform similar functions despite lacking sequence homology, indicates that the core function of these synapses lies in their connectivity rather than in their specific protein sequences. Previous models have proposed that AIB-AFD electrical synapses contribute to avoidance behavior by forming part of an FLP-AIB-AFD circuit, with AIB serving as a hub neuron that integrates sensory inputs via chemical synapse with FLP and electrical synapse with AFD [98]. Our experiments using engineered synapses elucidate the functional importance of the AIB-AFD connection in tactile-dependent locomotion modulation. Moreover, the transfer of signaling molecules such as Ca^{2+} and cGMP through electrical synapses likely facilitates the coordination of neuronal activity across distinct networks [99–102]. Although the precise mechanisms require further investigation, our study demonstrates that AIB-AFD electrical synapses are essential for incorporating tactile context into locomotion modulation.

In summary, our findings reveal an unexpected role for AFD in tactile-dependent behavioral plasticity. They show that specialized molecular mechanisms (including distinct cGMP signaling pathways, diverse guanylyl cyclases, and functional electrical synapse connectivity) enable AFD sensory neurons to perform dual functions: detect stimuli and support context-dependent plasticity. This work expands our understanding of sensory neuron versatility supporting adaptive behavior in animals.

Methods

Nematode Strains and Growth

C. elegans strains were maintained under standard conditions at 20°C on nematode growth medium (NGM) agar plates [103]. These plates were seeded with *E. coli* OP50 lawns. Strains used in this study are listed in **Table 1**.

Strain	Genotype	Citation
PR694	<i>tax-2(p694) I</i>	[106]
PR678	<i>tax-4(p678) III</i>	[65]
FX31482	<i>cng-2(tm4267) IV</i>	[107]
VC40261	<i>cng-3(gk541751) IV</i>	[108]
VC20290	<i>cng-4(gk195496) IV</i>	[108]
IK800	<i>gcy-8(oy44) IV.</i>	[43]
VC30137	<i>gcy-18(gk423024) IV</i>	[108]
VC20451	<i>gcy-12(gk142661) II</i>	[108]
BJH3481	<i>gcy-18(gk423024) IV, pekSi609[gcy-18p::gcy-18(cDNA)::sl2::mNeonGreen::unc-54 3'utr, vha-6p::mScarlet, hygR(+)] ttTi5605 II</i>	This study
BJH3527	<i>pekSi655[srtx-1Bp::cng-3(cDNA)::sl2::mNeonGreen::unc-54 3'utr, vha-9p::mScarlet, hygR(+)] ttTi5605 II; cng-3(gk541751) IV</i>	This study
BJH3537	<i>tax-2(p694) I; pekSi665[srtx-1Bp::tax2(cDNA)::sl2::mNeonGreen::unc-54 3'utr, vha-9p::mScarlet, hygR(+)] ttTi5605 II</i>	This study
BJH3539	<i>pekSi667 [gcy-18p::cng-3(cDNA)::sl2::mNeonGreen::unc-54 3'utr, vha-9p::mScarlet, hygR(+)] ttTi5605 II; cng-3(gk541751) IV</i>	This study
GN112	<i>pgls2[gcy-8p::TU#813, gcy-8p::TU#814, unc-122p::GFP, gcy-8p::mCherry, gcy-8p::gfp, ttx-3p::gfp]</i>	[71]
LX1024	<i>kcc-3(ok228) II</i>	[37]
PR767	<i>ttx-1(p767) V</i>	[38]
JN578	<i>pels578 [npr-9p::casp1 + npr-9p::venus + unc-122p::mCherry] (genetically ablated AIB)</i>	[74]
ZB2551	<i>mec-10(tm1552) X</i>	[72]
CB1611	<i>mec-4(e1611) X</i>	[109]
FG927	<i>inx-1(tm6662) X.</i>	[110]
FG0614	<i>inx-4 (ok2373) V</i>	[80]
RB1792	<i>inx-7(ok2319) IV</i>	[110]
RB2051	<i>inx-10(ok2714) V</i>	[110]
FX01896	<i>inx-19(tm1896) I</i>	[111]
BJH2919	<i>inx-7(ok2319) IV; inx-10(ok2714) V</i>	This study
BJH1162	<i>pekEx302 [srtx-1Bp::cx36::sl2::mNeonGreen, inx-1p::cx36::sl2::mKate, Cbr-unc-119(+), unc-122p::GFP, neoR(+)] unc-119(ed3) III; inx-7(ok2319) IV; inx-10(ok2714) V</i>	This study

Table 1.

List of *C. elegans* strains used in this study.

Single Copy Insertion of Transgenes

Mos1-mediated transgene insertion methods were used to generate transgenic animals carrying single-copy transgenes [104,105]. Briefly, worms carrying the *ttTi5605 Mos1* site on chromosome II were injected with a mixture of plasmids encoding the transgene of interest (40 ng/μl) and Cas9 protein plus guide RNA specific to the *Mos1* site (60 ng/μl). To aid in selection of successful insertions, the inserted transgene also included markers *vha-9p::mScarlet* and *rps-Op::hygR*, which confer red fluorescence in the gut and hygromycin resistance, respectively.

Transgenes and Germline Transformation

Strains carrying extrachromosomal arrays were generated by microinjecting various plasmids together with co-injection markers. The co-injection markers included *unc-112p::gfp* (coelomocyte, 30 ng/μl), *rpl-28p::neoR* (20 ng/μl), and *Cbr-unc-119* (20 ng/μl). The blank vector pBluescript was used as filler DNA to bring the final total DNA concentration to 100 ng/μl. A detailed list of the plasmids used to create transgenic worms in this study is provided in **Table 2**.

Fabrication of PDMS Microfluidic Chambers

Microfluidic chambers were designed in AutoCAD (S1. File), and the resulting CAD designs were used to fabricate SU8-Silicon molds with a height of 100 μm using photolithography (FlowJem, NO, Canada). These molds were then used to fabricate microfluidic chambers with various physical features using soft lithography with polydimethyl siloxane (PDMS, Sylgard 184, Dow Corning, MI). Specifically, PDMS prepolymer was poured onto the SU8-silicon molds and cured at 80°C for 2 hours. Once cured, the PDMS layer was carefully lifted away from the silicon wafer, and a 2 mm skin biopsy punch (Acuderm Inc., FL) was used to create inlet and outlet holes. To assemble the device, the PDMS layer was placed onto a microscope slide (1 mm thick), and the assembly was treated in a plasma cleaner (PDC-32G, Harrick Scientific, NY) for 1 minute. Distilled water was added to the plasma-treated devices within 5 minutes. Before loading the worms, the devices were thoroughly rinsed with M9 buffer.

Behavior Analysis

Worms were age-synchronized for behavioral assays by transferring 40–50 young adult worms onto an NGM plate seeded with OP50 and allowing them to lay eggs for 2 hours before removal. The plates were then kept at room temperature (20 ± 2°C) for 72 hours, allowing the eggs to develop into young adults. To wash the synchronized adults off the plate, worms were gently transferred into Eppendorf tubes using glass Pasteur pipettes. After allowing the worms to settle at the bottom of the tube (~1 minute), the supernatant was removed and replaced with fresh M9 buffer. This washing step was repeated three times to remove residual OP50. Next, approximately 30–70 worms were loaded into the chamber inlet using a 20 μl pipette and allowed to explore the chamber freely for 60 minutes before recording. Videos were captured for 10 seconds at 7.5 frames per second using a Basler acA2440 camera (Basler, PA) equipped with an AF Micro-NIKKOR 60 mm lens (Nikon, NY). Behavioral data were extracted from the videos using WormLab software (MBF Bioscience, VT) through individual worm tracking. Only those worms that were already within the assay area at the start of the recording were analyzed, and any worms that entered later were excluded. Crawling speed was determined by the total distance traveled by a worm during the 10 second recording. A turn was defined as a change in crawling direction of 15° or more. Reversals were defined as sudden movement in the opposite direction for at least 5 frames (~0.6 seconds). Worms were considered idle if their absolute velocity (distance traveled between frames) was below 20 μm/s for more than 10 frames (~1.3 seconds).

Name	Construct	Notes
BJP-MR16	<i>gcy-18p::gcy-18(mini-gene)::sl2::mNeonGreen::rab-3 3'utr; vha-9p::mScarlet::tbb-2 3'utr; rps-0p::hygR</i>	The <i>gcy-18</i> mini-gene includes all exons and introns 11, 12, and 13.
BJP-MR65	<i>srtx-1Bp::cng-3(cDNA)::sl2::mNeonGreen::rab-3 3'utr; vha-9p::mScarlet::tbb-2 3'utr; rps-0p::hygR</i>	
BJP-MR72	<i>srtx-1Bp::tax2(cDNA)::sl2::mNeonGreen::rab-3 3'utr; vha-9p::mScarlet::tbb-2 3'utr; rps-0p::hygR</i>	
BJP-MR93	<i>gcy-18p::cng-3(cDNA)::sl2::mNeonGreen::rab-3 3'utr; vha-9p::mScarlet::tbb-2 3'utr; rps-0p::hygR</i>	
BJP-MR96	<i>inx-1p::cx36::sl2::mKate::rab-3 3'utr</i>	
BJP-MR97	<i>srtx-1bp::cx36::sl2::mKate::rab-3 3'utr</i>	

Table 2.

List of plasmids used to create transgenic *C. elegans* strains.

Statistical Analysis

Behavior measurements were calculated for each individual chamber, and these chamber-specific values were then used to generate the final mean values shown in the bar plots. The sample size was defined as the number (approximately 25) of chambers analyzed per experiment. An unpaired Student's t-test with Bonferroni correction was used to compare average speeds between uniform and binary chambers. To quantify the relative difference between groups, we calculated the percent change for each value in the uniform chambers relative to the mean of value from the binary chambers. Specifically, we subtracted the mean of the values from the binary chamber from each individual value in the uniform chamber. The result was then divided by the mean of the binary chamber and multiplied by 100 to express the difference as a percentage. For comparisons among multiple strains, we conducted one-way ANOVA followed by a Tukey-Kramer post hoc test on all possible pairwise comparisons. All statistical analyses were performed using R.

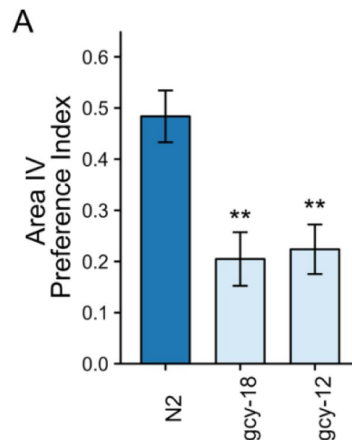
Acknowledgements

This research was supported by NIH grants R01NS109476 and R01NS115974 to JB. MR was supported by T32GM136534 and F31NS129545. The authors thank the *Caenorhabditis elegans* Genetics Center (funded by the NIH Office of Research Infrastructure Programs P40 OD010440) for providing worm strains. We also thank Lin Zhang and Yan Liu for their support in constructing DNA plasmids and generating transgenic worm strains. Special thanks to Melissa Woo (Mercer Island High School), Tabitha Ngo, and Sahithi Challam (Interlake High School) for their assistance with worm maintenance. We appreciate the members of the Bai lab for their critical reading of the manuscript. Additionally, we acknowledge the multiple mentorship programs at Fred Hutchinson Cancer Center, including the SURP Undergraduate Researchers and Pathways Undergraduate Researchers programs, supported in part by the Cancer Center Support Grant P30 CA015704, for fostering relationships with high school and undergraduate trainees who contributed to this project.

Suppl. Figure 1.

Worms lacking either *gcy-18* or *gcy-12* failed to identify their preferred area in microfluidic chambers.

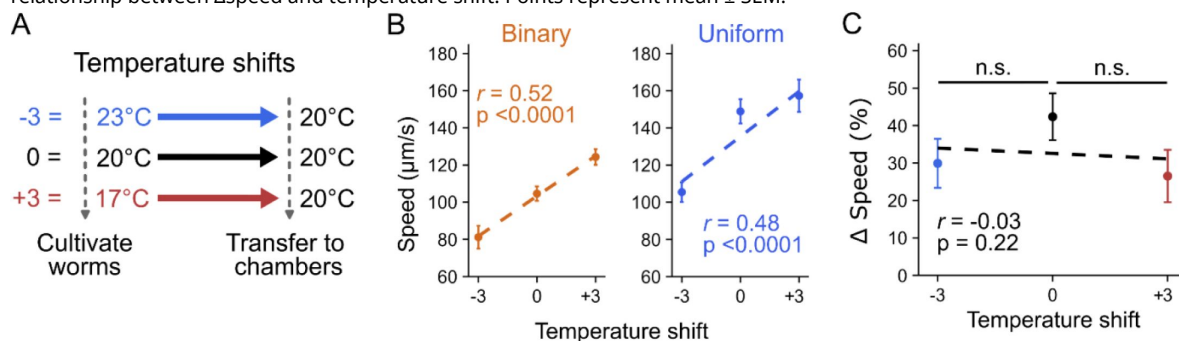
Preference index values of wild type N2 worms, *gcy-18* mutants, and *gcy-12* mutants were determined as described by Han *et al.* [52]. Wild type worms exhibit a strong preference for a specific zone (zone IV, as defined by Han *et al.* [52]), whereas both *gcy-18* and *gcy-12* mutants show significantly reduced preference compared to wild type worms. Bars represent mean $\pm$ SEM. Statistical comparisons were performed using one-way ANOVA followed by a Dunnett's test for multiple comparisons (** $p < 0.01$).



Suppl. figure 2.

Temperature shifts do not affect tactile-dependent modulation.

(A) Schematic of experimental design: Worms were reared at three different temperatures (17°C, 20°C, and 23°C) and then transferred to 20°C for 1 hour during the behavioral assay. (B) Locomotion speed of wild type N2 worms in the binary (orange) and uniform (blue) chambers after experiencing temperature shifts. Linear regression analysis reveals a significant positive correlation between rearing temperature and locomotion speed in both chambers. (C) Differences in speed (Δ speed) between worms experiencing different temperature shifts. Linear regression significance test shows no significant relationship between Δ speed and temperature shift. Points represent mean $\pm$ SEM.



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Reviewer #1 (Public review):

Summary:

In this manuscript, Rosero and Bai examined how the well-known thermosensory neuron in *C. elegans*, AFD, regulates context-dependent locomotory behavior based on the tactile experience. Here they show that AFD uses discrete cGMP signalling molecules and

independent of its dendritic sensory endings regulates this locomotory behavior. The authors also show here that AFD's connection to one of the hub interneurons, AIB, through gap junction/electrical synapses, is necessary and sufficient for the regulation of this context-dependent locomotion modulation.

Strengths:

This is an interesting paper showcasing how a sensory neuron in *C. elegans* can employ a distinct set of molecular strategies and different physical parts to regulate a completely distinct set of behaviors, which were not been shown to be regulated by AFD before. The experiments were well performed and the results are clear. However, there are some questions about the mechanism of this regulation. This reviewer thinks that the authors should address these concerns before the final published version of this manuscript.

Weaknesses:

(1) The authors argued about the role of prior exposure to different physical contexts which might be responsible for the difference in their locomotory behaviour. However, the worms in the binary chamber (with both non-uniformly sized and spaced pillars) experienced both sets of pillars for one hour prior to the assay and they were also free to move between two sets of environments during the assay. So, this is not completely a switch between two different types of tactile barriers (or not completely restricted to prior experience), but rather a difference between experiencing a more complex environment vs a simple uniform environment. They should rephrase their findings. To strictly argue about the prior experience, the authors need to somehow restrict the worms from entering the uniform assay zone during the 1hr training period.

(2) The authors here argued that the sensory endings of AFD are not required for this novel role of AFD in context-dependent locomotion modulation. However, *gcy-18* has been shown to be exclusively localized to the ciliated sensory endings of AFD and even misexpression of *GCY-18* in other sensory neurons also leads to localizations in sensory endings (Nguyen et. al., 2014 and Takeishi et. al., 2016). They should check whether *gcy-18* or *tax-2* gets mislocalized in *kcc-3* or *tax-1* mutants.

(3) *MEC-10* was shown to be required for physical space preference through its action in FLP and not the TRNs (PMID: 28349862). Since FLP is involved in harsh touch sensation while TRNs are involved in gentle touch sensation, which are the neuron types responsible for tactile sensation in the assay arena? Does *mec-10* rescue in TRNs rescue the phenotype in the current paper?

(4) The authors mention that the most direct link between TRNs and AFD is through AIB, but as far as I understand, there are no reports to suggest synapses between TRNs and AIB. However, FLP and AIB are connected through both chemical and electrical synapses, which would make more sense as per their *mec-10* data. (the authors mentioned about the FLP-AIB-AFD circuit in their discussion but talked about TRNs as the sensory modality). *mec-10* rescue experiment in TRNs would clarify this ambiguity.

(5) Do *inx-7* or *inx-10* rescue in AFD and AIB using cell-specific promoters rescue the behaviour?

(6) How Guanylyl cyclase *gcy-18* function is related to the electrical synapse activity between AFD and AIB? Is AFD downstream or upstream of AIB in this context?

<https://doi.org/10.7554/eLife.106496.1.sa2>

Reviewer #2 (Public review):

Summary:

The goal of the study was to uncover the mechanisms mediating tactile-context-dependent locomotion modulation in *C. elegans*, which represents an interesting model of behavioral plasticity. Starting from a candidate genetic screen focusing on guanylate cyclase (GCY) mutants, the authors identified the AFD-specific *gcy-18* gene as essential for tactile-context-dependent locomotion modulation. AFD is primarily characterized as a thermo-sensory neuron. However, key thermosensory transduction genes and the sensory ending structure of AFD were shown here to be dispensable for tactile-context locomotion modulation. AFD actuates tactile-context locomotion modulation via the cell-autonomous actions of GCY-18 and the CNG-3 cyclic nucleotide-gated channel, and via AFD's connection with AIB interneurons through electrical synapses. This represents a potentially relevant synaptic connection linking AFD to the mechanosensory-behavior circuit.

Strengths:

- (1) The fact that AFD mediates tactile-context locomotion modulation is new, rather surprising, and interesting.
- (2) The authors have combined a very clever microfluidic-based behavioral assay with a large set of genetic manipulations to dissect the molecular and cellular pathways involved. Rescue experiments with single-copy transgenes are very convincing.
- (3) The study is very clearly written, and figures are nicely illustrated with diagrams that effectively convey the authors' interpretation.

Weaknesses:

- (1) Whereas GCY-18 in AFD and the AFD-AIB synaptic connection clearly play a role in tactile-context locomotion modulation, whether and how they actually modulate the mechanosensory circuit and/or locomotion circuit remains unclear. The possibility of non-synaptic communication linking mechanosensory neurons and AFD (in either direction) was not explored. Thus, in the end, we have not learned much about what GCY-18 and the AFD-AIB module are doing to actuate tactile context-dependent locomotion modulation.
- (2) The authors only focused on speed readout, and we don't know if the many behavioral parameters that are modulated by tactile context are also under the control of AFD-mediated modulation.
- (3) The AFD-AIB gap junction reconstruction experiment was conducted in an innexin double mutant background, in which the whole nervous system's functioning might be severely impaired, and its results should be interpreted with this limitation in mind.

<https://doi.org/10.7554/eLife.106496.1.sa1>

Reviewer #3 (Public review):

Summary:

Rosero and Bai report an unconventional role of AFD neurons in mediating tactile-dependent locomotion modulation, independent of their well-established thermosensory function. They partially elucidate the signaling mechanisms underlying this AFD-dependent behavioral modulation. The regulation does not require the sensory dendritic endings of AFD but rather the AFD neurons themselves. This process involves a distinct set of cGMP signaling proteins

and CNG channel subunits separate from those involved in thermosensation or thermotaxis. Furthermore, the authors demonstrate that AIB interneurons connect AFD to mechanosensory circuits through electrical synapses. They conclude that, beyond its primary function in thermosensation, AFD contributes to context-dependent neuroplasticity and behavioral modulation via broader circuit connectivity.

While the discovery of multifunctionality in AFD is not entirely unexpected, given the limited number of neurons in *C. elegans* (302 in total), the molecular and cellular mechanisms underlying this AFD-dependent behavioral modulation, as revealed in this study, provide valuable insights into the field.

Strengths:

- (1) The authors uncover a novel role of AFD neurons in mediating tactile-dependent locomotion modulation, distinct from their well-established thermosensory function.
- (2) They provide partial insights into the signaling mechanisms underlying this AFD-dependent behavioral modulation.
- (3) The neural behavior assays utilizing two types of microfluidic chambers (uniform and binary chambers) are innovative and well-designed.
- (4) By comparing AFD's role in locomotion modulation to its thermosensory function throughout the study, the authors present strong evidence supporting these as two independent functions of AFD.
- (5) The finding that AFD contributes to context-dependent behavioral modulation is significant, further reinforcing the growing evidence that individual neurons can serve multiple functions through broader circuit connectivity.

Weaknesses:

- (1) Limited Behavioral Assays: The study relies solely on neural behavior assays conducted using two types of microfluidic chambers (uniform and binary chambers) to assess context-dependent locomotion modulation. No additional behavioral assays were performed. To strengthen the conclusions, the authors should validate their findings using an independent method, at the very least by testing AFD-ablated animals and *gcy-18* mutants with a second behavioral approach.
- (2) Clarity in Behavioral Assay Methodology: The methodology for conducting the behavioral assays is unclear. It appears that worms were free to move between the exploration and assay zones, with no control over the duration each worm spent in either zone. This lack of regulation may introduce variability in tactile experience across individuals, potentially affecting the reproducibility and quantitateness of the method. The authors should clarify whether and how they accounted for this variability.
- (3) Potential Developmental and Behavioral Confounds in Mutant Analysis: Several neuronal mutant strains were used in this study, yet the effects of these mutations on development and general behavior (e.g., movement ability) were not discussed. Although young adult worms were used for behavioral assays, were they at similar biological ages? To rule out confounding factors, locomotion assays assessing movement ability should be conducted (see reference PMID 25561524).
- (4) Definition and Baseline Measurements for Locomotion Categories: The finding that *tax-4* and *kcc-3* contribute to basal locomotion but not to context-dependent locomotion modulation is intriguing. The authors argue that distinct mechanisms regulate these two processes; however, the study does not clearly define the concepts of "basal locomotion" and

"context-dependent locomotion," nor does it provide baseline measurements. A clear definition and baseline data are needed to support this conclusion.

<https://doi.org/10.7554/eLife.106496.1.sa0>